FACULTY OF ENGINEERING, BUILT ENVIRONMENT AND INFORMATION TECHNOLOGY

BACHELOR IN INFORMATION TECHNOLOGY (HONOURS)
BACHELOR IN COMPUTER SCIENCE (HONOURS)
(WEEKEND (PART-TIME) & ODL)

BIT2233/BTL2333/BCL2233 COMPUTER ARCHITECTURE

CONTINUOUS ASSESSMENT 30%:
Assignment Topic: Computer Architecture Analysis Based on Professional or Industry Context

DUE DATE SUBMISSION: 25 MARCH 2026, 11:59PM

STUDENT NAME	
STUDENT ID	

DECLARATION OF INTEGRITY:

I hereby declare that this lab assessment submission is my own independent work and does not contain plagiarized content, unauthorized assistance, or any form of academic dishonesty. I confirm that I have adhered to the academic integrity policies outlined by the university.

I understand that if any form of academic dishonesty, including plagiarism, falsification of results, or unauthorized collaboration, is detected in this submission, I may face disciplinary actions. This may include, but is not limited to, receiving a failing grade for this assessment or further academic penalties as determined by the university's academic integrity committee.

By signing below, I acknowledge my understanding and agreement with this declaration.

Student's Signature: _____ Date: _____

CLO ASSESSED

Analyse architecture design considerations including I/O organisation and system integration trade-offs based on real-world scenarios. (C4, PLO2)

TABLE OF SPECIFICATION (ToS)

Task Component	Assessment Description	CLO	PLO	Bloom Level	Marks	%
Part A	Professional/Industry Scenario Identification	CLO3	PLO2	C3	15	15%
Part B	Architecture Components Analysis (CPU, Memory, I/O, Data Flow)	CLO3	PLO2	C3–C4	25	25%
Part C	I/O Organisation & System Integration Trade-off Analysis	CLO3	PLO2	**C4 (dominant)**	35	35%
Part D	Architecture Recommendation & Justification	CLO3	PLO2	C4	25	25%

Cognitive Distribution Summary

Assessment Component	CLO	PLO	Bloom
System architecture identification	CLO3	PLO2	C3
I/O organisation analysis	CLO3	PLO2	C4
Trade-off evaluation	CLO3	PLO2	C4
Architecture recommendation	CLO3	PLO2	C4

SLT JUSTIFICATION

Task Activity	Learning Mode	Bloom	Estimated SLT
Understand assignment & choose professional scenario	NF2F	C3	3 hrs
Research system architecture	NF2F	C3	5 hrs
Architecture diagram preparation	NF2F	C3	4 hrs
I/O organisation & trade-off analysis	NF2F	C4	8 hrs
Develop recommendations & justification	NF2F	C4	5 hrs
Report writing & refinement	NF2F	C3	4 hrs
Lecturer consultation / online discussion	F2F (live session)	Support	1 hr

Instruction Checklist

 This assignment is **individual work**.

 Use appropriate technical terms and logical justification where required.

 Submit the assignment via the **Learning Management System (LMS) - SPARK**.

 Upload a single file in **PDF format**.

 Content Requirements

- Professional or industry scenario clearly described
- Architecture components identified and explained
- Architecture diagram included
- I/O organisation analysis provided
- System integration trade-offs analysed (not just described)
- Justified architecture recommendation included

 Format Requirements

- Typed document (except diagrams, which may be hand-drawn and scanned clearly).
- Font: Times New Roman / Arial
- Font size: 11 or 12
- Line spacing: 1.5
- Include:
    - Student Name
    - Student ID
    - Programme
    - Course Code
    - Lecturer's Name

AI Usage Policy (If Applicable)

If AI tools (e.g., ChatGPT, Copilot) are used:

- Students must clearly state how the tool was used
- AI cannot replace original coding work
- Students must understand and explain submitted code

Failure to declare AI usage may be considered misconduct.

Assessment Purpose

This assessment requires students to analyse computer architecture design considerations by relating theoretical concepts to real-world scenarios drawn from their own profession, working environment, or industry domain. Students will evaluate architecture decisions, I/O organisation, and system integration trade-offs relevant to practical computing systems.

PART A: Professional Context Identification (15 marks)

Students must:

- Describe their profession / working environment / industry sector
- Identify **ONE** computing system used in that context.

Examples:

- Banking system (finance industry)
- Hospital monitoring system
- Manufacturing automation system
- Education LMS platform
- Retail POS system
- Cloud-based collaboration tools

Students without work experience may choose:

- Future career field
- Publicly known system such as Smart campus system / e-commerce server / IoT monitoring system

PART B: Architecture Component Analysis (25 marks)

Students analyse:

- CPU role
- Memory usage
- Storage
- I/O devices
- Data flow

Deliverable:

- Architecture diagram.

PART C: I/O Organisation & Integration Trade-Off (35 marks)

Students analyse:

Examples:

- Polling vs interrupt
- Network I/O handling
- Real-time vs batch processing
- Performance vs cost
- Security vs usability

Requirement: justification based on scenario.

PART D: Architecture Recommendation (25 marks)

Students propose:

- Improved architecture or design suggestion
- Reasoning based on analysis.

- END OF ASSIGNMENT QUESTION -

Marking Scheme

Assessment: Architecture Design Analysis Based on Professional Context

Total Marks = 100
Weightage = 30%
CLO measured = CLO3
PLO mapped = PLO2 (Cognitive / Problem Analysis Skills)

Criteria	Marks	Extended Abstract (80–100)	Relational (65–79)	Multistructural (50–64)	Unistructural (40–49)	Prestructural (<40)	CLO	Bloom	Mark Allocation
Professional Context & System Selection	15	Realistic scenario strongly aligned with profession; clear relevance	Appropriate context with good explanation	Context described but limited relevance	Basic or unclear scenario	Poor or missing	CLO3	C3	
Architecture Component Analysis	25	Accurate, integrated explanation of CPU, memory, I/O and data flow	Mostly correct with clear understanding	Several components described separately	Limited explanation	Incorrect concepts	CLO3	C3–C4	
I/O Organisation & Trade-off Analysis	35	Deep comparative analysis with strong justification	Clear analytical comparison	Basic evaluation without depth	Minimal analysis	No meaningful analysis	CLO3	**C4**	
Architecture Recommendation & Justification	25	Logical recommendation strongly supported by analysis	Appropriate recommendation with reasoning	Basic suggestion with limited justification	Weak justification	Missing or unsupported	CLO3	C4	

Overall Feedback

First Marker:

_____
_____
_____
_____

Total Mark: _____

Signature	Date

Second Marker:

_____
_____
_____
_____

Total Mark: _____

Signature	Date